How accurate is the constant-denominator approximation in Hierarchical Correlation Reconstruction neurons?

A pre-registered numerical study

Przemysław Spyra
AGH University of Krakow
pspyra@agh.edu.pl

Abstract

Hierarchical Correlation Reconstruction (HCR) neurons model a joint density in an orthonormal polynomial basis and can propagate a whole density through a neuron. For that step the conditional density's input-dependent denominator $D(y)$ is replaced by a constant, which gives $p_{\mathrm{CD}} = \mathrm{E}_q[N]/\mathrm{E}_q[D]$ instead of $p_{\mathrm{EU}} = \mathrm{E}_q[N/D]$. This paper characterises that approximation. An exact identity writes the difference as a covariance under the belief q and shows that p_{CD} is exact propagation of the belief tilted by D. A numerical study frozen in a version-controlled pre-registration then measures the approximation in 2,900 GPU jobs, on synthetic copulas with up to 32 inputs, polynomial degree up to 6, and chains of up to four neurons. The difference is small at low input dependence and for narrow beliefs: with a flat belief and two inputs, the total-variation difference reaches 1 % at Spearman correlation 0.192, and with a narrow belief it stays at or below 0.0032 wherever it is defined. The tighter limit is that the exact reference stops existing: D turns negative, and the onset falls from 0.435 at two inputs to 0.007 at 32 (pair basis, $n = 10^7$). Where both effects can be measured, the effect of calibrating negative densities exceeds the denominator effect at 68 % of points. Of 15 pre-registered hypotheses, 5 are supported, 2 inconclusive and 8 not supported. Those not supported include all three about transfer to non-Gaussian dependence, compounding over depth, and all three about training under the fixed, untuned settings used here.

1 Introduction

A Hierarchical Correlation Reconstruction (HCR) neuron stores a joint density of its inputs and output as coefficients of an orthonormal polynomial basis [11, 12]. Because the neuron holds a density and not only a function, it can do more than a conventional unit: it can be evaluated in any direction, and it can take a density over its inputs, a *belief*, and return a density over its output. The second operation needs an integral of a ratio of two polynomials. In its density-propagation formula (Eq. 10 of [12]) HCR avoids that integral by treating the ratio's denominator as a constant and normalising afterwards, so that propagation reduces to a few inner products of coefficient vectors. The source paper states this openly as an approximation. It also says that the calibration needed to keep polynomial densities nonnegative is mostly neglected in neural-network use, and that the resulting approximations should be compensated during training.

How large is the error, and on what does it depend? We are not aware of a numerical answer. This paper gives one, for one neuron family and synthetic ground truths, as an independent characterisation and not as a critique. HCR's constant denominator buys a large saving: in the forward-pass microbenchmark of Section 6, one calibrated, sample-based alternative costs about 647 times the energy at one setting (not a like-for-like comparison). The study therefore asks when that saving comes at a small price and when it does not, and it keeps two sources of error apart: replacing the denominator, and leaving densities uncalibrated.

The findings are as follows.

- **An exact form of the error.** The difference between p_{CD} and p_{EU} is a covariance under the belief, divided by $\mathrm{E}_q[D]$. Equivalently, p_{CD} propagates the D-tilted belief exactly. This yields a total-variation bound in terms of $\mathrm{sd}_q(D)/|\mathrm{E}_q[D]|$ and explains the special cases: narrow beliefs, flat beliefs and the data belief (Section 3).
- **The difference itself is often modest.** For a flat belief at two inputs, d_{denom} reaches 1 % at $\rho_S = 0.192$. For a narrow belief it never exceeds 0.0032 (pair basis).
- **The dominant limit is negativity of D.** The negativity onset falls strongly with the number of inputs (H1a, H1b *supported*), with degree (H2a *supported*) and from pair to triple basis (H2b *supported*). Beyond the onset the reference p_{EU} is undefined, so a blank in our tables means "cannot be measured", never "small".
- **Sampling noise moves the onset at large k.** At $n = 10^7$ the triple basis at $k \geq 16$ has negative D already at independence. A closed-form variance explains the scale of this effect.
- **Calibration usually matters more than the denominator**, at 68 % of the points where both can be measured. This is consistent with the source paper's emphasis, though neither approximation dominates everywhere.
- **Gaussian results do not transfer** to other copula families or random correlation structures (H3a–c *not supported*).
- **Depth and training did not behave as hypothesised.** Error did not compound (H5a *not supported*), but this is consistent with the target mixing towards uniform, and layers beyond the first have nearly flat D, so the test is weak. Training under fixed, untuned settings did not meet the pre-registered criteria for reducing the error (H5b–d *not supported*); this is a statement about these settings only.

All outcomes are listed in Table 1.

2 Background

2.1 HCR neurons

HCR works on variables normalised to $[0, 1]$, for instance by their marginal distribution functions, so the object modelled is a copula density [41, 51]. Let $f_0 \equiv 1$ and $f_p(u) = \sqrt{2p+1}\, P_p(2u-1)$ be the orthonormal shifted Legendre polynomials on $[0, 1]$. For a multi-index $\mathbf{j}$ write $f_{\mathbf{j}}(y) = \prod_l f_{j_l}(y_l)$. A neuron with output x and inputs $y = (y_1, \ldots, y_k)$ models

$$N(x, y) = \sum_{i=0}^{m} f_i(x)\, A_i(y), \qquad A_i(y) = \sum_{\mathbf{j}} a_{i\mathbf{j}}\, f_{\mathbf{j}}(y), \tag{1}$$

with degree at most m per variable. The coefficients are mixed moments, estimated from a sample by $\hat{a}_{i\mathbf{j}} = \frac{1}{n}\sum_t f_i(x_t) f_{\mathbf{j}}(y_t)$ [12]. Since all coefficients up to degree m in $k+1$ variables are too many for large k, a neuron keeps only multi-indices with few nonzero entries. We use two index families: *pair*, with at most two nonzero entries among the $k+1$ variables, and *triple*, with at most three. Row $i = 0$ thus has up to two (pair) or three (triple) active inputs, and rows $i > 0$ one less. With o the order, the number of coefficients is $\sum_{r \leq o} \binom{k+1}{r} m^r$: for $k = 32$ and $m = 4$, 8,581 (pair) and 357,765 (triple).

Integrating over x removes all rows $i > 0$, so

$$D(y) := \int_0^1 N(x, y)\, dx = A_0(y) \tag{2}$$

is the neuron's model of the input density, and N/D its conditional density of x given y.

2.2 Density propagation and the constant denominator

Given a belief q on the inputs, exact propagation under the model is

$$p_{\mathrm{EU}}(x) = \mathrm{E}_q\left[\frac{N(x,y)}{D(y)}\right]. \tag{3}$$

The subscript stands for *exact, uncalibrated*. The HCR formula [12, Eq. 10] replaces this by

$$p_{\mathrm{CD}}(x) = \frac{\mathrm{E}_q[N(x,y)]}{\mathrm{E}_q[D(y)]} = \frac{\sum_i f_i(x)\sum_{\mathbf{j}} a_{i\mathbf{j}} b_{\mathbf{j}}}{\sum_{\mathbf{j}} a_{0\mathbf{j}} b_{\mathbf{j}}}, \qquad b_{\mathbf{j}} = \mathrm{E}_q[f_{\mathbf{j}}(y)]. \tag{4}$$

The right-hand form follows from (1) and costs one matrix–vector product. It sees q only through the moments $b_{\mathbf{j}}$. Note that $\int p_{\mathrm{CD}}\,dx = 1$ by construction whenever $\mathrm{E}_q[D] \neq 0$. We call this the *constant-denominator* approximation.

2.3 Negativity and calibration

A truncated orthogonal expansion of a nonnegative density need not be nonnegative. Orthogonal-series density estimation has a long history [5, 17, 31, 50], and general corrections that turn an estimator, including an orthogonal-series one, into a bona fide density are known [21, 23]. The HCR paper proposes a calibration: a map such as $\max(\rho, 0.1)$ or a softplus, followed by division by the integral [12, Sec. II-A]. In neural use, it states, the issue can usually be neglected, because the resulting approximations should be compensated during training; log-likelihood evaluation, however, needs a positive density and hence calibration [12, Sec. IV].

p_{EU} needs D to stay positive on the support of q, since $1/D$ is not integrable across a simple sign change. So there are two approximations in play, and they are distinct. Constant-denominator propagation is what (4) does. Skipping calibration is what lets N, D and the outputs be signed. Our design separates them by computing p_{EU} where it exists, a calibrated reference p_{EC}, and the ground truth (Section 4).

2.4 Related work

HCR. HCR grew out of polynomial density fitting [9, 11] and has been applied to time series [10], yield curves [13], bid–ask spreads [16], income data [14] and redshift prediction [15]. Legendre-based KAN layers with an HCR reading are studied in [53]. We found no prior numerical study of the constant-denominator step. Outside the HCR line, shifted-Legendre projection estimators of copula densities are developed in [42], and a Legendre multiwavelet variant in [6].

Ratios of expectations. That the mean of a ratio differs from the ratio of means by a covariance term is classical in sampling theory. It goes back to Hartley and Ross [25]; Schochet et al. [49] state the identity in the form used here, Meng [39] studies the related absolute bias ratio, and ratio estimators are treated in Cochran [7]. Our Proposition 1 applies that identity pointwise in x; the identity itself is not new. The same ratio-of-averages structure, with its bias, appears in self-normalised importance sampling [4, 45], and second-order delta-method expansions of a mean of a ratio are standard [44].

Propagating uncertainty through networks. Most methods that push distributions through networks keep a parametric form, often Gaussian. Examples are probabilistic backpropagation [27], natural-parameter networks [56], lightweight probabilistic networks based on assumed density filtering [22], deterministic moment approximation [58], analytic Gaussian inference [24] and covariance propagation [57]; see [28] for the unscented transform and [40] for expectation propagation. HCR differs by keeping a polynomial density, which is less restrictive in shape but can be negative. Boyen and Koller [2] show for stochastic processes that the error of approximate belief states contracts as the process mixes, which is relevant to our depth results.

Pre-registration. Fixing questions and analyses before seeing outcomes separates prediction from postdiction [43]. Its use in machine learning has been advocated in [20] and piloted in a NeurIPS workshop [1]. See [34, 47] for wider concerns and [48, 52] on computational reproducibility.

3 Analysis

Fix q and write $h_x(y) = N(x, y)/D(y)$. Throughout this section $D \neq 0$ q-almost surely, h_x is q-integrable, and $\mathrm{E}_q[D] \neq 0$.

3.1 The identity

Proposition 1. *For every x,*

$$p_{\mathrm{CD}}(x) - p_{\mathrm{EU}}(x) = \frac{\mathrm{Cov}_q(h_x(y),\, D(y))}{\mathrm{E}_q[D]}. \tag{5}$$

Proof. Since $N = h_x D$, we have $\mathrm{E}_q[N] = \mathrm{E}_q[h_x D] = \mathrm{E}_q[h_x]\mathrm{E}_q[D] + \mathrm{Cov}_q(h_x, D)$. Divide by $\mathrm{E}_q[D]$ and use $p_{\mathrm{EU}} = \mathrm{E}_q[h_x]$. ($D$ is bounded, so the covariance exists.) □

Proposition 2. *Let $\tilde{q}$ be the signed measure $d\tilde{q} = (D/\mathrm{E}_q[D])\, dq$, of total mass 1. Then*

$$p_{\mathrm{CD}}(x) = \int h_x(y)\, d\tilde{q}(y). \tag{6}$$

If $D/\mathrm{E}_q[D] > 0$ on the support of q, then $\tilde{q}$ is a probability measure, and Eq. (4) is exact propagation of the belief $\tilde{q}$ instead of q.

Proof. $\int h_x\, d\tilde{q} = \mathrm{E}_q[h_x D]/\mathrm{E}_q[D] = \mathrm{E}_q[N]/\mathrm{E}_q[D]$. □

So the constant denominator does not distort the conditional; it changes the belief. It reweights q towards inputs that the neuron considers likely, as if multiplying the belief by the neuron's own prior.

Corollary 1. *Pointwise, $|p_{\mathrm{CD}}(x) - p_{\mathrm{EU}}(x)| \leq \mathrm{sd}_q(h_x)\,\mathrm{sd}_q(D)/|\mathrm{E}_q[D]|$. Let C_1 be the supremum over y in the support of q of $\int |h_x(y)|\, dx$. Then $C_1 \geq 1$, with equality when the conditional is nonnegative, and*

$$\mathrm{TV}(p_{\mathrm{CD}}, p_{\mathrm{EU}}) = \frac{1}{2}\int |p_{\mathrm{CD}} - p_{\mathrm{EU}}|\, dx \;\leq\; \frac{C_1}{2}\,\mathrm{E}_q\left|\frac{D}{\mathrm{E}_q[D]} - 1\right| \;\leq\; \frac{C_1}{2}\,\frac{\mathrm{sd}_q(D)}{|\mathrm{E}_q[D]|}. \tag{7}$$

Proof. The first bound is Cauchy–Schwarz applied to (5). For the second, (6) gives $p_{\mathrm{CD}} - p_{\mathrm{EU}} = \int h_x\,(D/\mathrm{E}_q[D] - 1)\, dq$; take absolute values, integrate over x, exchange the integrals, and apply Jensen's inequality. $C_1 \geq 1$ follows from $\int h_x\, dx = 1$. □

The ratio $\mathrm{sd}_q(D)/|\mathrm{E}_q[D]|$ is a cheap diagnostic: it needs only D and the belief, not the x direction. On the production data the ratio of d_{denom} to $\frac{1}{2}\mathrm{sd}_q(D)/|\mathrm{E}_q[D]|$ never exceeded 0.86 over 149,880 defined rows, with median 0.037 (post hoc), so the bound held without C_1 and is usually loose. That is expected, because it ignores how weakly h_x and D may be correlated.

3.2 When the error vanishes

Equation (5) vanishes for all x exactly when h_x and D are uncorrelated under q for all x. Three cases matter.

D constant on the support of q. This covers a point belief, where both forms give the conditional at that point, and independent inputs at population level, where $D \equiv 1$. Narrow beliefs approximate this case, with error controlled by (7).

Flat belief. Let q be uniform, so $b_{\mathbf{j}} = 0$ for $\mathbf{j} \neq 0$. Then $\mathrm{E}_q[D] = a_{00} = 1$ and $p_{\mathrm{CD}}(x) = \sum_i a_{i0} f_i(x)$, the model's x-marginal, in any basis and regardless of how x depends on y. In the view of Proposition 2, $\tilde{q} = D$: the flat belief is replaced by the neuron's own input density, and the law of total probability returns the marginal. Also $\mathrm{sd}_q(D) = \chi_D$, where

$$\chi_D = \Big(\int (D-1)^2 \, dy \Big)^{1/2} = \Big(\sum_{\mathbf{j} \neq 0} a_{0\mathbf{j}}^2 \Big)^{1/2} \tag{8}$$

measures the distance of D from flat. The production data confirm this: the median of $\mathrm{sd}_q(D)/\chi_D$ under the flat belief is 1.00.

Data belief. Let q be the true input density with population coefficients, so $b_{\mathbf{j}} = a_{0\mathbf{j}}$ on the indices of row 0. Then $\mathrm{E}_q[D] = 1 + \chi_D^2$. In the pair basis, each row $i > 0$ contains only single-input terms, and with uniform marginals those moments of q vanish, so with a uniform x-marginal $p_{\mathrm{CD}} \equiv 1$ exactly, which is also the correct answer. In the triple basis, rows $i > 0$ contain two-input terms, and p_{CD} deviates: it is a real error, not noise. The script `paper/check_identity.py` verifies (5), (6), (7) and these special cases on a sample case.

3.3 Why D turns negative, and why k matters

For Legendre polynomials $f_1(u) f_1(v)$ has expectation $12 \, \mathrm{E}[(u - \frac{1}{2})(v - \frac{1}{2})] = \rho_S$ on a copula, so $a_{0,(1,1)} = \rho_S$ for each input pair exactly, and $\chi_D \geq \rho_S \sqrt{k(k-1)/2}$. With equicorrelated inputs, χ_D thus grows like k at fixed ρ_S. In the sweep, $\chi_D / (\rho_S \sqrt{k(k-1)/2})$ lies between 1.04 and 1.05 (SUM, pair, $0.05 \leq \rho_S \leq 0.2$).

D is largest in magnitude at corners of the cube, where $f_1 = \pm\sqrt{3}$. As a heuristic, keep only the degree-one pair terms and take a corner with balanced signs $s_l = \pm 1$, $\sum_l s_l = 0$. Then

$$D \approx 1 + 3\rho_S \sum_{l<l'} s_l s_{l'} = 1 - \tfrac{3}{2} \rho_S k, \tag{9}$$

which reaches zero at $\rho_S \approx 2/(3k)$. This neglects higher degrees, so it predicts only the $1/k$ scaling. The measured population onsets give $k\rho_S^*$ between 0.82 and 1.26 for the pair basis, and between 0.36 and 0.37 for triple at $k \geq 4$.

Sampling noise adds to this. Under independence $\hat{a}_{0\mathbf{j}}$ has variance $1/n$ and the estimates are uncorrelated, so $\mathrm{Var}\hat{D}(y) = \frac{1}{n} \sum_{\mathbf{j} \neq 0} f_{\mathbf{j}}(y)^2$. At a corner $f_p^2 = 2p + 1$; with $T = \sum_{p=1}^{m} (2p+1) = m(m+2)$,

$$\sigma_{\mathrm{c}}^2 = \frac{1}{n} \Big(kT + \binom{k}{2} T^2 \Big) \text{ (pair)}, \qquad \sigma_{\mathrm{c}}^2 = \frac{1}{n} \Big(kT + \binom{k}{2} T^2 + \binom{k}{3} T^3 \Big) \text{ (triple)}. \tag{10}$$

Similarly $\mathrm{E}\,\hat{\chi}_D^2 = |B_D|/n$, with $|B_D|$ the number of non-constant row-0 coefficients. With $m = 4$ and $n = 10^7$, σ_{c} is 0.17 for pair at $k = 32$, 0.88 for triple at $k = 16$ and 2.6 at $k = 32$. Since the minimum runs over many corners, $1 - \min \hat{D}$ is a multiple z of σ_{c}; in our data z grows from 1.0 at $k = 2$ to 4.4 at $k = 32$ (Table 9). Hence, writing $K = n\sigma_{\mathrm{c}}^2$ for the bracket in (10), keeping $\hat{D} > 0$ at independence needs roughly $n \gtrsim z^2 K$. For the triple basis with $k = 32$ this is far beyond $n = 10^7$.

4 Experimental design

The design, hypotheses, decision rules, grids and budget rule were frozen in a version-controlled pre-registration (file `PREREGISTRATION.md`, commit `6a98646`) before any production job. It is a git record, not an entry in a public registry. A pilot, run before the freeze to measure throughput only (seed index 1000), is excluded from all results. The Deviations section of the file is empty. Analyses marked *post hoc* were not pre-registered and do not enter any verdict. Full definitions are in Appendix A.

4.1 Ground truth

Output and inputs are uniform; dependence comes from a $(k+1)$-dimensional copula. The Gaussian sweep has equicorrelated inputs with correlation $r = 2\sin(\pi\rho_S/6)$, so ρ_S is the Spearman correlation between inputs. Two types differ in how x couples to the inputs. In **SUM** each input has correlation $c = \min(0.5,\, 0.8\sqrt{(1+(k-1)r)/k})$ with x. In **CONTRAST** half the inputs have $+c$ and half $-c$, with $c = \min(0.5,\, 0.8\sqrt{(1-r)/k})$. The non-Gaussian experiments use t_3 and t_{10} copulas [8] in SUM, CONTRAST and exchangeable (EXCH) forms, plus exchangeable Clayton and Gumbel copulas [38], all at matched Kendall's τ of the inputs [33]. Archimedean samples use the mixture representation of Marshall and Olkin [37], now used to sample Archimedean copulas, with Kanter's method [29] for the positive stable variables. Random correlation matrices come from the LKJ distribution [32].

4.2 Beliefs

Beliefs are product distributions unless stated. **S2** is flat. **S3a** is Beta$(3,3)$ per input, **S3b** is Beta$(4,2)$, **S3c** alternates Beta$(4,2)$ and Beta$(2,4)$ across inputs (conflicting evidence) and **S3d** is the narrow Beta$(6,6)$. **S4** is the true input copula (data belief). **S1**, the neuron's own D, is computed but not primary. The primary beliefs are S2, S3c and S4. S2 is not the data distribution, so part of its error against the ground truth is mismatch between prior and data, not error of the neuron.

4.3 Compared densities and outcomes

For each belief, Monte Carlo with $M = 2^{20}$ samples in $R = 16$ replicate blocks gives: p_{CD}; p_{EU}, defined only if $D \geq 0.01$ on the support of q; p_{EC}, the conditional calibrated per input as $\max(N, \varepsilon)$ divided by its integral (192-node Gauss–Legendre, $\varepsilon \in \{0.1, 0.01, 0.001\}$, primary 0.01); and p_{G}, the analytic conditional of the ground truth averaged over q. Note that per-input calibration is one choice among several and is not the same as normalising a joint density.

Outcomes are total-variation distances $\mathrm{TV}(p,r) = \frac{1}{2}\int |p-r|$, computed on raw, never renormalised densities on an 801-node grid in normal score:

$$d_{\mathrm{denom}} = \mathrm{TV}(p_{\mathrm{CD}}, p_{\mathrm{EU}}), \qquad d_{\mathrm{neg}} = \mathrm{TV}(p_{\mathrm{EU}}, p_{\mathrm{EC}}), \qquad d_{\mathrm{model,uncal}} = \mathrm{TV}(p_{\mathrm{EU}}, p_{\mathrm{G}}),$$
$$d_{\mathrm{model,cal}} = \mathrm{TV}(p_{\mathrm{EC}}, p_{\mathrm{G}}), \qquad d_{\mathrm{total}} = \mathrm{TV}(p_{\mathrm{CD}}, p_{\mathrm{G}}), \qquad d_{\mathrm{dc}} = \mathrm{TV}(p_{\mathrm{CD}}, p_{\mathrm{EC}}).$$

Because densities can be signed, TV here is an L^1 distance with a factor $\frac{1}{2}$ and can exceed 1; it is a probability-like quantity only between genuine densities. We also record the relative variance error, the standardised mean error, $P(D<0)$ under uniform y, χ_D, and $\min D$ from a multi-start projected optimiser with 2,000 starts. This $\min D$ is an upper bound on the true minimum, so negativity onsets are conservative: true onsets can only be earlier.

4.4 Thresholds and uncertainty

Each seed re-estimates all coefficients. A threshold $\rho_S^*(c)$ is the first crossing of level c by the seed-median curve, by linear interpolation over the run of defined points from $\rho_S = 0$. If the curve

becomes undefined first, or c is not reached by $\rho_S = 0.9$, the entry is "—". The negativity onset is the first zero crossing of seed-median $\min D$, and 0 if it is already negative at $\rho_S = 0$. "Undefined from" is the first grid point at which at least 50 % of seeds have p_{EU} undefined. Confidence intervals are percentile bootstraps over seeds [18, 19] ($B = 2000$), paired where conditions share seeds. Values below three times their Monte Carlo noise floor are flagged and reported, not subtracted.

A hypothesis evaluated over cells is *supported* if at least 2/3 of the evaluable cells are significant in the predicted direction, *not supported* if at least 2/3 are not, and *inconclusive* otherwise or with fewer than three evaluable cells. Some hypotheses have their own rule (Appendix D). In this paper "significant" is used only in this pre-registered sense.

4.5 Experiments

`gauss` crosses $k \in \{2, 4, 8, 16, 32\}$, both types, both bases, $m \in \{2, 4, 6\}$, $n = 10^7$, 50 seeds, and 19 values of ρ_S from 0 to 0.9. `population` repeats these cells with exact coefficients. `convergence` doubles M, n, and the two grid sizes on a subset. `families` ($k \leq 8$, 20 seeds) and `lkj` ($k \leq 8$, 200 matrices per setting) vary the ground truth. `network` builds chains (below). Altogether 2,900 jobs ran with 0 failures on NVIDIA A100-SXM4-40GB GPUs in float64 with TF32 disabled and deterministic algorithms, using 176.7 GPU-hours of job time (196.25 GPU-hours billed for everything, including validation and benchmarks).

4.6 Chains

The teacher is a Markov chain in normal scores. The first hidden variable depends on y through a Gaussian-copula neuron of the sweep; for $l \geq 2$, $s_{h_l} = a\, s_{h_{l-1}} + b\, s_{w_l} + \sqrt{1 - a^2 - b^2}\, e_l$ with $a = 0.6$, $b = 0.5$, fresh independent side inputs w_l and noise e_l. Neuron $l \geq 2$ is a triple-basis model of $(h_l; h_{l-1}, w_l)$, that is $k = 2$. Neuron 1 uses pair or triple. Depth is $L \leq 4$, with $k \in \{2, 4, 8\}$, $\rho_S \in \{0, 0.2, 0.4, 0.6\}$, $m = 4$ and 10 seeds. Estimation uses $n = 10^6$; training, validation and test splits have $2 \cdot 10^5$, 10^5 and 10^5 samples. Between layers the only normalisation is $c_0 = 1$, the first coefficient of the propagated polynomial; there is no re-mapping of variables to quantiles.

Training uses three losses: value MSE as in Eq. 7 of [12], calibrated negative log-likelihood (NLL), and end-to-end (e2e) squared L^2 distance between the p_{CD} output and the projected target, summed over S2, S3c and S4. It uses two optimisers, Adam [30] and L-BFGS [35], each with fixed, untuned settings (Appendix G); the best validation state is kept. A flat-D control starts from $a_{0\mathbf{j}} = 0$ for $\mathbf{j} \neq 0$.

5 Results

5.1 Q1: dependence on the number of inputs

H1a, H1b (*supported*, *supported*). The negativity onset falls strictly with k in all 4 population cells, and at $n = 10^7$ the difference between $k = 32$ and $k = 2$ has a confidence interval below zero in all 4 cells. For SUM with the pair basis the onset falls from 0.435 to 0.007, a difference with 95 % CI $[-0.429, -0.428]$ (Table 2, Figures 2 and 3).

Table 2 needs careful reading. The 1 % threshold of d_{denom} also falls with k, from 0.192 ($k = 2$) to 0.043 ($k = 16$) in the pair basis, while at $k = 32$ the entry is empty. Empty does not mean that the error stays small. It means that p_{EU} became undefined (from $\rho_S = 0.010$) before the curve reached 1 %. Likewise, in Table 2 the 5 % level is reached only at $k = 2$ in the pair basis, because at $k \geq 4$ the reference disappears first. For the triple basis at $k \geq 16$ the reference is undefined already at $\rho_S = 0$ under S2.

	claim	verdict	evidence
H1a	population onset falls with k	*supported*	4/4 cells strictly decreasing
H1b	same at $n = 10^7$	*supported*	4/4; SUM/pair 0.435 ($k = 2$) vs 0.007 ($k = 32$)
H1c	d_{denom} one power law of χ_D	*inconclusive*	within factor 2: CONTRAST/pair 0.52, CONTRAST/triple 0.48, SUM/pair 0.55, SUM/triple 1.00
H2a	higher m, earlier onset	*supported*	20/20
H2b	triple earlier than pair (population)	*supported*	8/8
H2b	triple earlier than pair ($n = 10^7$)	*supported*	8/8
H2c	d_{denom} depends weakly on m	*inconclusive*	60 % of 221 points; supported needs ≥ 80 %
H3a	onset within ± 0.05 of Gaussian	*not supported*	21/48 CIs inside, 27 entirely outside
H3b	d_{denom} within factor 2 of Gaussian	*not supported*	73/236 CIs inside, 156 entirely outside
H3c	Gaussian power law predicts LKJ	*not supported*	pair 39 %, triple 34 % of 272 matrices; $k = 2$ only
H4a	seed spread exceeds MC error	*supported*	93 % of 114 points; median ratio 2.1
H4b	doubling changes TV $< \max(3\,\text{SE}, 10^{-3})$	*not supported*	5,460 violations, all under $n \times 2$
H5a	d_{total} grows from $L = 1$ to 4	*not supported*	11/72; 51 significantly below 1
H5b	training reduces d_{denom}	*not supported*	5/18; 1 opposite
H5c	training reduces $P(D < 0)$	*not supported*	24/78; 19 opposite
H5d	e2e training reduces held-out d_{total}	*not supported*	27/192; 84 opposite

Table 1: All pre-registered outcomes, with the verdict words of the pre-registration. H2b has two pre-registered evaluations, so 15 hypotheses occupy 16 rows. Fractions count significant cells among evaluable cells; "opposite" counts cells whose confidence interval lies entirely on the unpredicted side. H4b is also test M13 of the validation suite (Appendix F).

The sampled onset is earlier than the population onset at large k: 0.007 against 0.039 (pair, $k = 32$), and 0.000 against 0.023 (triple, $k = 16$). This is the noise of Section 3: at $\rho_S = 0$ the observed $1 - \min D$ follows $z\,\sigma_c$ with z between 1.0 and 4.4 in both bases, and χ_D matches $\sqrt{|B_D|/n}$ within 2 % (post hoc, Table 9).

H1c (*inconclusive*). If d_{denom} under S2 were a single function of χ_D, one power law would fit each (type, basis). The share of points within a factor 2 was 0.55 (SUM/pair), 0.52 (CONTRAST/pair), 1.00 (SUM/triple) and 0.48 (CONTRAST/triple). By the pre-registered rule these cells are respectively *inconclusive*, *inconclusive*, *supported* and *not supported*, and the overall verdict is *inconclusive* (Table 11). The pre-registration gives this rule per (type, basis) and does not say how the cells combine. The overall label comes from the analysis code frozen at the same commit, which requires all cells to agree. Counting a *supported* cell as significant under the general aggregation rule would instead give 1 of 4, that is *not supported*; the cell verdicts are the informative result. The pattern is consistent with the identity: χ_D fixes $\text{sd}_q(D)$, but the covariance also depends on how h_x varies, which changes with k and type through the coupling c.

5.2 Q2: dependence on the basis

H2a (*supported*). A higher degree gives an earlier onset in 20 of 20 population cells (Table 12).

H2b (*supported* with population coefficients, *supported* at $n = 10^7$). The triple basis has an earlier onset than pair in 8 of 8 and 8 of 8 cells. The two bases coincide for D at $k = 2$, which is why $k = 2$ is not in this hypothesis.

H2c (*inconclusive*). d_{denom} varied by at most a factor 1.5 across $m \in \{2, 4, 6\}$ at 60 % of 221 points; the rule for *supported* is 80 % and for *not supported* 50 %. Degree thus moves the onset consistently, while its effect on d_{denom} where defined is, by this rule, neither clearly weak nor clearly strong.

basis	k	$d_{\text{denom}} \geq 1\,\%$	$d_{\text{denom}} \geq 5\,\%$	onset, $n = 10^7$	onset, pop.	undefined from
pair	2	0.192	0.359	0.435	0.437	0.450
pair	4	0.102	—	0.201	0.204	0.200
pair	8	0.067	—	0.123	0.129	0.150
pair	16	0.043	—	0.059	0.074	0.075
pair	32	—	—	0.007	0.039	0.010
triple	2	0.085	—	0.435	0.437	0.450
triple	4	0.020	—	0.085	0.094	0.100
triple	8	0.009	—	0.016	0.045	0.020
triple	16	—	—	0.000	0.023	0.000
triple	32	—	—	0.000	0.011	0.000

Table 2: Thresholds in ρ_S for type SUM, flat belief S2, $m = 4$, seed-median curves, $n = 10^7$ except the population onset. "—" means the level was not reached before p_{EU} became undefined; it never means that the error is small. "undefined from" is the first grid value at which most seeds have p_{EU} undefined. Onsets are upper bounds. CONTRAST and all confidence intervals are in Table 6.

family	cells	onset − Gaussian (min … max)	CI inside	CI outside	median ratio
clayton	6	-0.179 … +0.006	2	4	5.34
gumbel	6	-0.128 … -0.013	1	5	16.60
t10	18	-0.102 … +0.036	15	3	2.08
t3	18	-0.201 … -0.016	3	15	7.22

Table 3: Comparison with the Gaussian copula at matched Kendall's τ ($m = 4$, pair and triple, $k \in \{2, 4, 8\}$, $n = 10^7$, 20 seeds). Columns: range of onset differences; number of 95 % CIs inside and entirely outside ± 0.05; median ratio of d_{denom} to Gaussian under S2.

5.3 Q3: dependence on the ground truth

H3a (*not supported*). The rule requires every onset difference from the Gaussian to have a CI inside ± 0.05. Of 48 comparisons, 21 were inside and 27 entirely outside (Table 3, Figure 4). Onsets were mostly earlier than Gaussian, with differences as low as -0.201 for t_3 and -0.179 for Clayton.

H3b (*not supported*). d_{denom} under S2 at matched τ fell within a factor 2 of the Gaussian, with its CI, at 73 of 236 points, and entirely outside at 156. Median ratios to the Gaussian were 2.08 (t_{10}), 5.34 (Clayton), 7.22 (t_3) and 16.60 (Gumbel).

A likely part of the t result is structural. A t copula is dependent at zero correlation, so matched $\rho_S = 0$ is not independence: at $k = 2$ under the pair basis χ_D at $\rho_S = 0$ is 0.30 for t_3 and 0.09 for t_{10}, against 0.0015 for the Gaussian, and t_3 already has $\min D$ =-0.67 at $k = 4$ (post hoc). Tail dependence loads the corners, where the basis is largest. We did not test this explanation further.

H3c (*not supported*). For LKJ matrices the Gaussian power law predicted d_{denom} within a factor 2 for 39 % (pair) and 34 % (triple) of 272 defined matrices, against a rule of at least 80 % for *supported* and below 50 % for *not supported*. This rests on $k = 2$ only: at $k = 4$ and $k = 8$ the number of defined matrices was 0 and 0 (Table 13).

5.4 Q4: uncertainty and convergence

H4a (*supported*). At 93 % of 114 headline points the across-seed standard deviation of d_{denom} exceeded the Monte Carlo standard error, with median ratio 2.1. Estimation noise, not integration noise, sets the uncertainty.

H4b (*not supported*). The hypothesis, also test M13, requires doubling M, n, the s-nodes or the x-nodes to change every reported TV by less than $\max(3\,\text{SE}, 10^{-3})$. There were 5,460

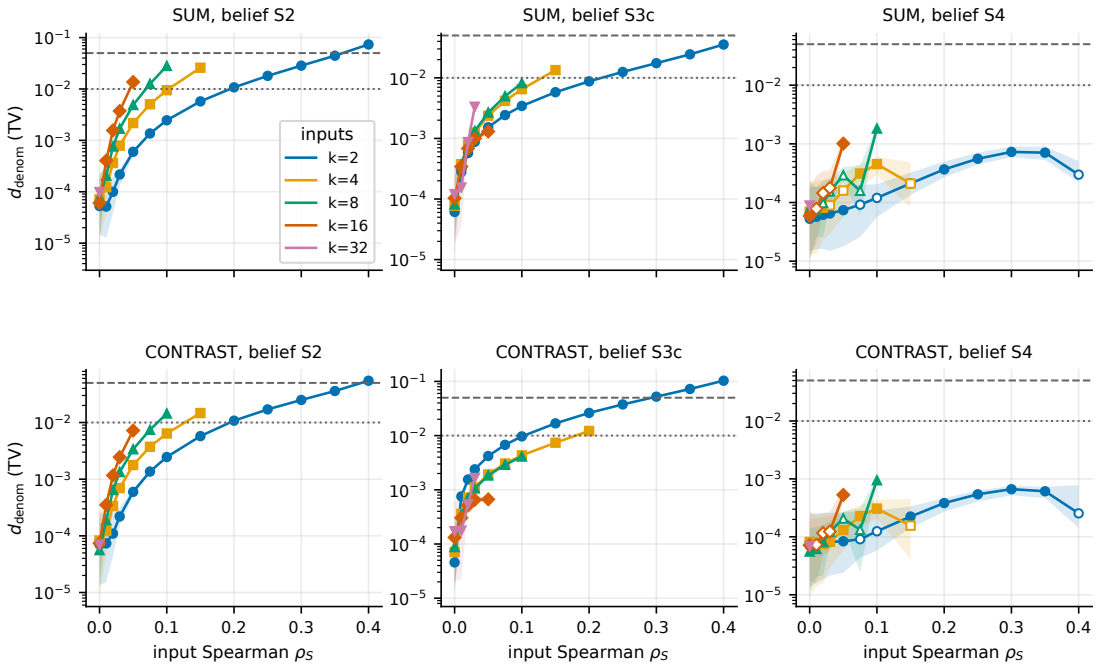


Figure 1: d_{denom} against input Spearman correlation ρ_S for the pair basis, $m = 4$, $n = 10^7$, $\varepsilon = 0.01$, beliefs S2, S3c and S4. Lines are seed medians and bands 2.5–97.5 % seed intervals. A curve ends where most seeds have p_{EU} undefined; the gap carries no information about the error. Hollow markers are below three times the Monte Carlo noise floor. Dotted and dashed lines mark 1 % and 5 %.

violations among 1,165,059 paired values (0.47 %). All of them are under $n \times 2$, where they are 5,460 of 291,258 (1.9 %), and they occur at $k = 8$ (498) and $k = 32$ (4,962). The worst ratio of change to tolerance is 36.2 for $n \times 2$ and 0.86 over the three other variants (Table 14). Thus numerical integration is converged, while at large k the estimator is not: 93 % of violations are in the triple basis, and their median relative change is 2.8 % (post hoc). The largest absolute change, 8.3, is from 26.4 to 18.1, at $k = 32$, triple, $m = 6$, S3d; values of that size arise only for signed densities (Section 5.6).

5.5 Q5: depth and training

H5a (*not supported*). The ratio $d_{\text{total}}(L = 4)/d_{\text{total}}(L = 1)$ exceeded 1 significantly in 11 of 72 cells and was significantly below 1 in 51. Median ratios are 0.05 (S2), 0.06 (S3c) and 1.25 (S4) (Table 4, Figure 5).

This is not good news by itself. What is measured is that the target moves. Under S2 the variance of p_{G} rises towards the uniform value 1/12 with depth (from 0.0713 to 0.0829), and p_{CD} sits near it throughout, so d_{total} falls because the true output loses its memory of the belief. In a Gaussian chain with correlation a per layer, a Hermite mode of order r shrinks by a^r per layer, so a variance-type deviation (order 2) shrinks by $a^6 = 0.047$ over three layers, close to the S2 ratio. This is consistent with contraction of the error under a mixing Markov kernel, in the spirit of Boyen and Koller [2], but contraction is a hypothesis here, not something we tested. Moreover, the S4 median of 1.25 is not uniform: pair reaches 1.91 (8 of 12 cells significant), triple 0.04. The significant cells all have tiny errors: $d_{\text{total}}(L = 4) \leq 0.0012$, at most 2.43 times the noise floor (so below the flagging level), and 4 are at $\rho_S = 0$ (post hoc).

The chain also does not stress the denominator after layer one. Inputs (h_{l-1}, w_l) of later neurons are independent, so their D is nearly flat: median $\min D$ is 0.97, lowest value 0.93. H5a therefore does not test compounding of denominator error across layers with correlated inputs. Only neuron 1 has dependent inputs.

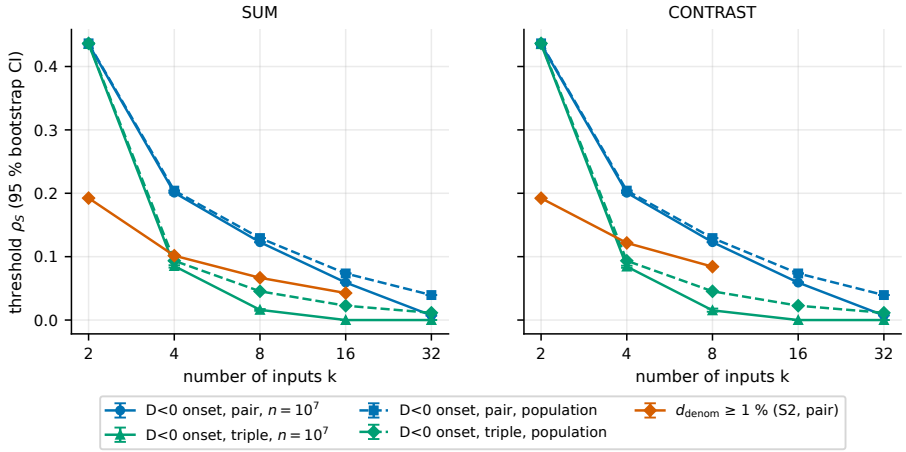


Figure 2: Negativity onsets for both bases, at $n = 10^7$ (solid) and with population coefficients (dashed), and the 1 % threshold of d_{denom} (S2, pair), against k, for $m = 4$. Error bars are 95 % bootstrap CIs and are often smaller than the markers. Missing points mean the level was not reached before p_{EU} became undefined.

belief	basis	$L = 1$	$L = 2$	$L = 3$	$L = 4$	Var p_{G}, $L = 1$	$L = 4$	ratio	above/below/all
S2	pair	0.0648	0.0211	0.0071	0.0024	0.0713	0.0829	0.05	1/9/12
S2	triple	0.0653	0.0213	0.0071	0.0025	0.0713	0.0829	0.05	1/9/12
S3c	pair	0.0516	0.0111	0.0043	0.0018	0.0642	0.0826	0.04	0/12/12
S3c	triple	0.0126	0.0071	0.0030	0.0015	0.0642	0.0826	0.10	0/12/12
S4	pair	0.0005	0.0010	0.0010	0.0011	0.0833	0.0833	1.91	8/0/12
S4	triple	0.0748	0.0252	0.0086	0.0031	0.0833	0.0833	0.04	1/9/12

Table 4: Depth results by belief and neuron-1 basis, medians over k, $\rho_S > 0$ and seeds. Columns: d_{total} at each layer; variance of p_{G} at $L = 1$ and $L = 4$; median H5a ratio over cells including $\rho_S = 0$; numbers of H5a cells significantly above 1, significantly below 1, and total. The breakdown is post hoc; the verdict uses all 72 cells.

H5b (*not supported*). Training reduced d_{denom} of neuron 1 significantly in 5 of 18 evaluable cells, with 1 opposite. Only $k = 2$ is evaluable, as d_{denom} is mostly undefined at larger k. By loss (post hoc): NLL 4 of 6, value 1 of 4, e2e 0 of 8.

H5c (*not supported*). Training reduced $P(D < 0)$ in 24 of 78 cells, with 19 opposite. Here the loss matters: NLL 21 of 26 (4 opposite), value 3 of 26 (8 opposite), e2e 0 of 26 (7 opposite). NLL lowers the negative mass (median change of $P(D < 0)$ -0.055 with Adam, -0.043 with L-BFGS, where it was positive) while deepening the minimum (min D falls by 7.3 and 8.1).

H5d (*not supported*). End-to-end training reduced held-out d_{total} significantly in 27 of 192 cells, with 84 opposite. The count rises with k (0, 6, 21 of 64 each at $k = 2, 4, 8$) and is larger in the triple basis (23 against 4 of 96). At $L = 1$, median d_{total} falls on the training beliefs (S2 by 0.036 with Adam and 0.031 with L-BFGS, S3c by 0.021 and 0.021), and rises on held-out beliefs (by 0.006 and 0.025), so the optimiser did learn, and part of what it learned did not generalise across beliefs (post hoc).

Three facts limit these verdicts (Table 15, Figure 6). First, 54 % (Adam) and 50 % (L-BFGS) of the e2e runs at $L = 4$ kept the initial state, with median coefficient change zero; this is consistent with the depth-4 target being nearly uniform and the loss being at noise level. These runs are the ones used for H5b and H5c, so 8 of 18 H5b cells and 26 of 78 H5c cells barely measure training. Second, optimiser settings were fixed and untuned, and data were limited. Third, d_{denom} can be evaluated only at $k = 2$. We therefore conclude that training did not reduce these errors under these settings. The data do not show that training cannot compensate for the approximations, and the source paper's expectation that training compensates for neglected

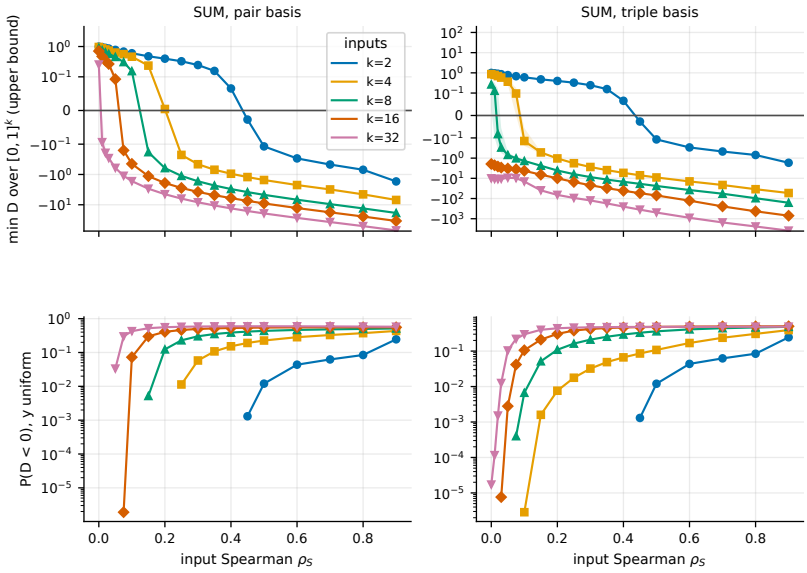


Figure 3: Negativity of D for type SUM, $m = 4$, $n = 10^7$, seed medians. Top: $\min D$ over $[0,1]^k$ on a symmetric logarithmic axis; the value is an upper bound on the true minimum. Bottom: $P(D < 0)$ under uniform y on a logarithmic axis; points with zero observed probability are not drawn.

calibration remains untested with more capacity, longer optimisation or a downstream task. There were no non-finite losses in the 2,880 training runs from direct estimates. The flat-D control starts with $d_{\text{denom}} = 0$; after NLL training it has median $P(D < 0) = 2.1\,\%$ and $\chi_D = 0.52$, so likelihood training moves a flat model into the negative regime.

5.6 Separating the error sources

This section is descriptive and post hoc, although all metrics were pre-registered. It uses $m = 4$ and $\varepsilon = 0.01$ unless stated.

Denominator against calibration. At 873 points where d_{denom} and d_{neg} both exist and at least one is above the noise floor, $d_{\text{neg}} > d_{\text{denom}}$ at 68 % of points (72 % for $\varepsilon = 0.1$, 67 % for $\varepsilon = 0.001$). By belief: S2 57 %, S3a 77 %, S3b 55 %, S3c 82 %, S3d 69 %, S4 69 %; the median ratio is largest under S4 (27.6). This fits the source paper's emphasis on calibration as the practical issue. These points all lie where D is positive on the support of q, however, so the comparison is silent where negativity is strongest.

Which output is closer to the truth? p_{CD} is closer to p_{G} than p_{EC} in 57 % of rows: 60 % where p_{EU} is defined and 55 % where it is not. The share depends strongly on belief, from 98 % (S4, pair) to 24 % (S3b, pair). So p_{CD} is not a uniformly worse output than the calibrated conditional, nor a uniformly better one. Clipping per input with a small ε is a blunt calibration, and these figures say nothing about better ones.

What S2 measures. Under S2, p_{CD} is the marginal (Section 3) and d_{total} is close to $\text{TV}(p_{\text{G}}, \text{uniform})$: the median absolute gap is 1.6×10^{-4}, the maximum 0.018. The S2 value of d_{total} thus measures mismatch between flat prior and truth more than any failing of the neuron. Figure 7 shows the components, and Table 10 gives values at $k = 2$.

Data belief. Under S4 the pair basis has median $d_{\text{total}} = 2.9 \times 10^{-4}$ and maximum 9.2×10^{-4}, as predicted. The triple basis has median 0.015 and maximum 0.28 ($k \geq 4$, $\rho_S > 0$ for the median).

Belief width and position. S3d, the narrow belief, has d_{denom} at most 0.0032 (pair). At $k = 16, 32$ it stays defined up to $\rho_S = 0.9$ and never reaches 1 %, so here "—" does mean small; at $k = 2, 4, 8$ it stops being defined at 0.600, 0.350 and 0.350 (Table 8). Width is not everything: the off-centre S3b reaches 1 % earlier than S2 at $k \geq 8$ (at $k = 8$, 0.014 against 0.067).

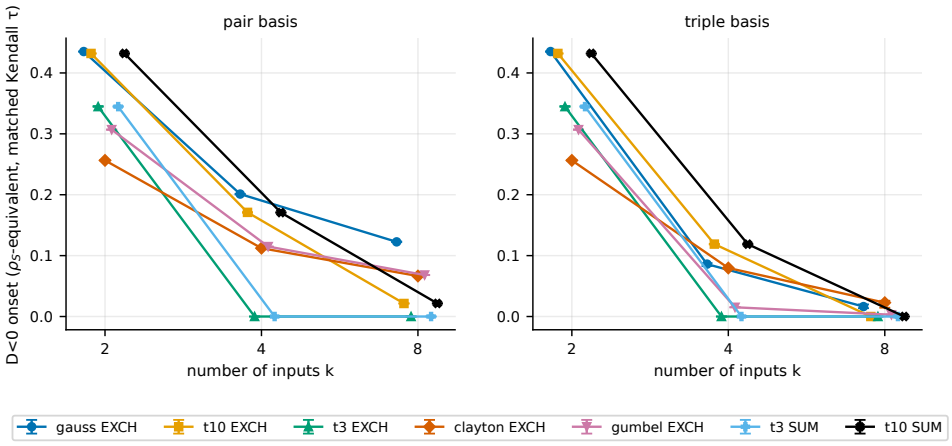


Figure 4: Negativity onset by copula family at matched Kendall's τ, expressed in ρ_S units, $m = 4$, $n = 10^7$, 20 seeds, with 95 % bootstrap CIs (mostly smaller than the markers). Points are shifted slightly in k for legibility. EXCH families and the t SUM forms are shown; t CONTRAST is in Table 3. An onset of 0 means $\min D < 0$ at $\rho_S = 0$.

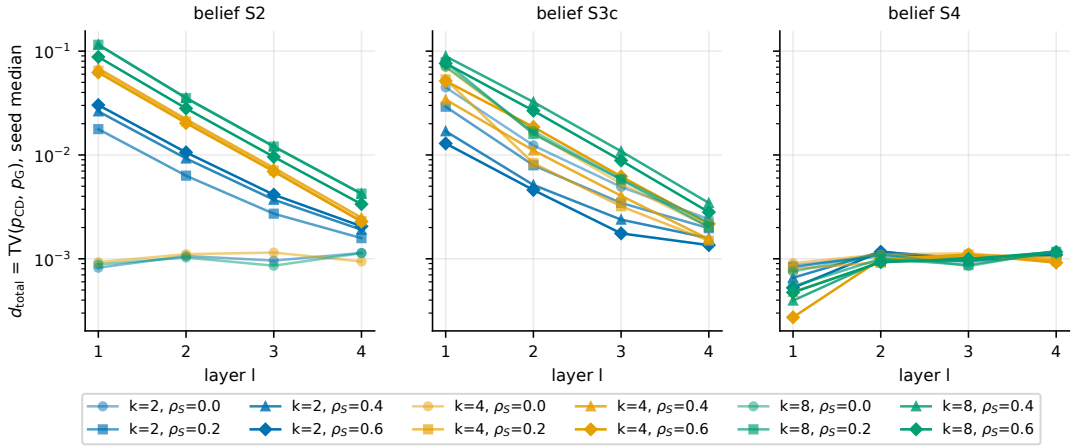


Figure 5: d_{total} by layer for chains with direct estimates: seed medians over 10 seeds, neuron 1 in the pair basis, $m = 4$, $n = 10^6$, for $k \in \{2, 4, 8\}$ and $\rho_S \in \{0, 0.2, 0.4, 0.6\}$. The decline under S2 and S3c reflects the target approaching the uniform density, not improved accuracy. Values near 10^{-3} are at Monte Carlo noise level.

Mean or variance? Among 181 points of the pair basis, the relative variance error exceeds the standardised mean error at 78 % (Figure 9). Under S2 the mean is almost untouched (at most 0.0020) while the variance moves by up to 13.5 %. Under S3c CONTRAST both move, up to 0.57 and 0.38.

p_{CD} *can also fail.* Equation (4) stays computable when p_{EU} is undefined, but not always meaningful. $\mathrm{E}_q[D]$ is a polynomial in the belief's moments and can be nonpositive. This happened in 22,297 of 399,000 rows (6,900 of 133,000 at $m = 4$), all in the triple basis, at $k \geq 8$, under the Beta beliefs, with a minimum of −1,449. With population coefficients it occurred in 2,230 of 39,900 rows, also from $k = 8$, so it is structural and not only noise. The pair basis stayed at $\mathrm{E}_q[D] \geq 0.71$. Relatedly, $d_{\text{total}} > 1$ in 3,245 rows (0.8 %), 95 % of them triple, with a largest value of 473: the output is then a strongly signed function, not a density.

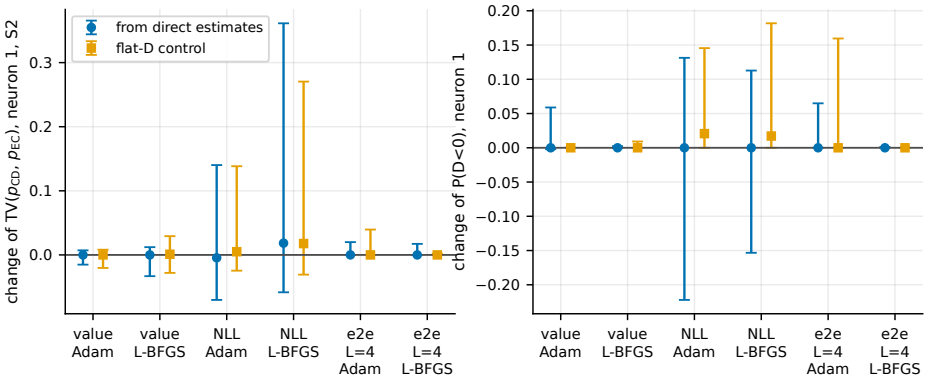


Figure 6: Effect of training on neuron 1 under S2. Left: change of $d_{\mathrm{dc}} = \mathrm{TV}(p_{\mathrm{CD}}, p_{\mathrm{EC}})$, shown because it exists everywhere; this is not the pre-registered d_{denom} of H5b. Right: change of $P(D < 0)$. Points are medians and bars 10–90 % ranges over all k, ρ_S, bases and seeds, from direct estimates and from the flat-D control. Runs labelled e2e are the $L = 4$ runs.

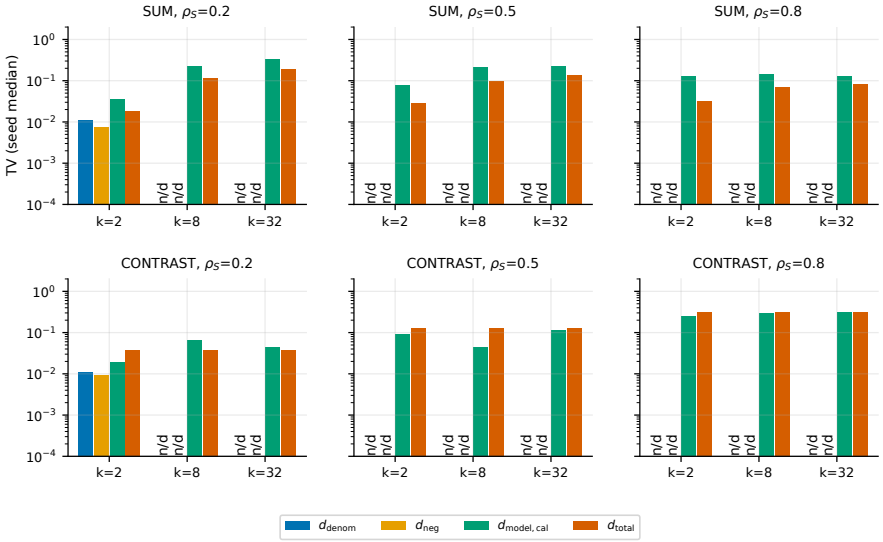


Figure 7: Error components under the flat belief S2, pair basis, $m = 4$, $n = 10^7$, $\varepsilon = 0.01$; seed medians at three correlations and three values of k. "n/d" marks quantities not defined because p_{EU} is undefined, not values near zero. d_{total} under S2 is dominated by the mismatch between the flat belief and the truth.

6 Time and energy

This section is a separate descriptive benchmark and never enters the error tables. Its purpose is to give the cost of what the approximation avoids. It compares one forward pass of four parameter-matched layers of width 64: HCR-CD (Eq. 4, pair, $m = 4$), HCR-CAL (calibrated conditional, $S = 64$ samples per belief, 192 x-nodes), an MLP layer with GELU, and a KAN layer [36] (grid 5, order 3). Settings are $k \in \{2, 8, 32\}$ and batch $B \in \{256, 16384\}$, in 6 runs (3 per precision) of 30 repeats of about 2 s, with NVML power sampling at 10 ms on one A100 type (commit `2d26201`). Table 5 shows one setting; all are in Appendix H.

At this setting HCR-CD uses 2.0 times the energy of the MLP layer in float32 and 1.9 times in float64, KAN uses 3.5 and 4.1 times that of HCR-CD, and HCR-CAL uses 647 and 828 times. We draw no general claim that HCR is more or less efficient than the other layers. The comparison is parameter-matched, forward only and inference-shaped, on one GPU type with clocks not locked, and all implementations are unoptimised. The layers compute different things: an MLP returns values, and HCR-CD returns a density. The factor for HCR-CAL reflects $S = 64$

layer	parameters	float32 ms	float32 J	float64 ms	float64 J
HCR-CD	39,232	0.484	0.137	0.666	0.203
HCR-CAL	39,232	308	88.7	580	168
MLP	39,265	0.289	0.0682	0.297	0.105
KAN	39,528	1.91	0.482	3.32	0.829

Table 5: One forward pass at $k = 8$, $B = 16384$, median over the 90 repeats of 3 runs per precision, energy including idle power. A descriptive microbenchmark of unoptimised code; see the caveats in the text. The layers do different jobs, and HCR-CAL uses $S = 64$ samples per belief, so rows are not like-for-like.

samples and many x-nodes per belief, a design choice, and not a like-for-like cost.

Measurement quality was acceptable for this purpose. The energy counter agrees with integrated sampled power at median ratio 0.999 (1st–99th percentile 0.953–1.061) over 4,320 repeats, which matters because built-in sensors have known sampling problems [3], including readings through `nvidia-smi` on the A100 that cover only part of the runtime [59]. Idle power dominates 8 of 48 rows. Run-to-run differences follow the physical GPU, with one GPU 5–6 % above the others, in two runs only. The float64 MLP at $k = 32$, $B = 16384$ draws 398 W, at the 400 W power limit, so 71 repeats ran at 97.5 % of the limit or above. Also, float64 is faster than float32 for the MLP at $k = 2$, $B = 16384$ (0.0667 ms against 0.145 ms), an effect of cuBLAS kernel selection that a separate probe confirmed. On reporting energy in machine learning, see [26].

7 Discussion

7.1 What the approximation does well

The formula is cheap, always normalised when $\mathrm{E}_q[D] \neq 0$, and needs only moments of the belief. It is exact for point beliefs and, in the pair basis, for the data belief at population level. It behaves well for narrow beliefs: $d_{\mathrm{denom}} \leq 0.0032$. It remains computable over most of the range where p_{EU} is undefined, and there it is closer to the truth than a per-input clipped conditional in 55 % of rows. Where both can be measured, the denominator effect is smaller than the calibration effect at most points. Within the limits below, these facts support its use as a fast propagation rule.

7.2 Where it needs care

Under a flat belief it returns the marginal whatever the dependence is, so it cannot be read as "the conditional averaged over ignorance". This follows directly from Proposition 2. With many inputs and positive dependence, D is negative over much of the cube and p_{EU} does not exist as a reference. With the triple basis at $k \geq 8$ under the Beta beliefs $\mathrm{E}_q[D]$ can itself be nonpositive, and the output is then not a density in any useful sense. Under the data belief, the triple basis deviates in a way the pair basis does not. The most extreme settings, such as the triple basis at $k = 32$, are stress tests: the source paper itself suggests restricting to pairwise dependencies and selecting the most valuable coefficients.

7.3 Guidance for users

1. *Check the denominator.* Compute $\mathrm{E}_q[D]$ and $\mathrm{sd}_q(D)/|\mathrm{E}_q[D]|$ for the beliefs of interest; both are cheap. Half of the ratio bounds d_{denom} up to C_1. A nonpositive or small $\mathrm{E}_q[D]$ should stop the propagation.

2. *Monitor* $\min D$. As a rule of thumb from Section 3, with equicorrelated inputs expect negativity near $\rho_S \sim 1/k$ for the pair basis at $m = 4$, and earlier in the triple basis or at higher degree.
3. *Prefer the pair basis with low degree* when k is large, unless higher-order dependence is needed. Use (10) to size n: the triple basis at $k \geq 16$ was noise-dominated even at $n = 10^7$.
4. *Do not transfer Gaussian thresholds* to heavy-tailed or asymmetric dependence (H3a–c). Our thresholds are not universal constants.
5. *Interpret a flat-belief output as a marginal.*

7.4 Limitations

The scope is one neuron family, a Legendre basis, degree $m \leq 6$, and synthetic copulas. There is no real-data task and no downstream accuracy, so we cannot say how much the measured differences matter in application. Specifically:

- The reference is available only where $D \geq 0.01$ on the support of q. This floor is a pre-registered convention and was not varied. Beyond it, we compare with p_{G} and p_{EC} only, so d_{denom} describes the easier regime.
- p_{EC} is per-input clipping, not the joint normalised calibration of [12]. A different calibration could change Section 5.6.
- TV on signed densities is an L^1 distance, not a probability bound.
- Thresholds lie on a grid; onsets are upper bounds, since the minimiser may miss deeper minima.
- The Gaussian main sweep has equicorrelated inputs, a special structure that favours large pair sums, and H3c addressed general matrices only at $k = 2$. The coupling c is a design choice that enters through the covariance in (5). Marginals are exactly uniform, whereas real pipelines use estimated quantiles.
- The basis keeps all coefficients of its family. Selecting coefficients, which the source paper suggests, would reduce both the noise of (10) and the corner sums, and was not tested.
- Confidence intervals are per cell, with no correction for multiplicity; verdicts rest on the pre-registered counting rules.
- The chain is contractive ($a = 0.6$ fixed), has $k = 2$ independent-input neurons after layer one, and no inter-layer re-normalisation to quantiles. The source paper discusses normalisation between layers, which we did not test.
- Training used one fixed setting per optimiser. e2e runs at $L = 4$ often kept the initial state.
- The source paper's value-propagation mode does not use Eq. 10, and this study says nothing about it.
- The pre-registration is a commit in a private history until the code is released. Commit timestamps alone are not third-party proof; a public registry would be stronger evidence.
- Several explanations offered here (tail dependence in Q3, mixing in Q5) are consistent with the data but were not tested.

8 Conclusion

The constant-denominator approximation in HCR neurons has a simple exact description: it propagates the belief tilted by D. Its error is controlled by how much D varies under the belief.

Numerically it is small for narrow beliefs and low dependence, and the larger practical limit is negativity of the modelled input density, which arrives near $\rho_S \sim 1/k$ and earlier with noise, higher degree, a richer basis or non-Gaussian dependence. Calibration usually moves the output more than the denominator does, which agrees with the emphasis of the source paper. Whether training compensates for either effect remains open: with fixed, untuned settings on synthetic chains it did not meet the pre-registered criteria. The most useful next steps are tests on real tasks, with chains whose inputs stay correlated, and with joint calibration.

Acknowledgements

[TODO: acknowledgements, subject to consent of those named; funding] [TODO: PLGrid grant acknowledgement]

Code and data availability

[TODO: code URL / DOI] Appendix J lists commits, seeds, hardware and software.

References

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A Full definitions

Support and floor. The support of a belief is $\{q \geq 10^{-6} \max q\}$. p_{EU} is *defined* for a row if no Monte Carlo sample in the support and no optimiser point in the support has $D < 0.01$.

TV. $\mathrm{TV}(p, r) = \frac{1}{2} \int |p - r|$ on 801 nodes in normal score $s \in [-8, 8]$ with a cubic sub-interval rule, on raw signed densities. Under a monotone change of variable TV does not change; the normal score resolves the sharp conditional densities at high correlation.

Variance and mean errors. $\mathrm{relvar} = (\mathrm{Var}_{\mathrm{CD}} - \mathrm{Var}_{\mathrm{ref}})/\mathrm{Var}_{\mathrm{ref}}$ in x, and $\mathrm{smean} = |\mathrm{Mean}_{\mathrm{CD}} - \mathrm{Mean}_{\mathrm{ref}}|/\mathrm{sd}_{\mathrm{ref}}$, with reference p_{EU} (primary) or p_{G}.

Noise. With R replicate blocks, $\mathrm{SE} = \mathrm{sd}(\mathrm{replicates})/\sqrt{R}$, and the noise floor is the mean replicate-difference L^1 divided by $\sqrt{R-1}$.

Seed-median curve. Undefined seed values count as $+\infty$; a grid point is defined if at least 50 % of seeds are defined. Thresholds are per seed too, and bootstrap CIs resample seeds; "not reached" counts as $+\infty$ in the percentile.

B Grids

$\rho_S \in \{0, 0.01, 0.02, 0.03, 0.05, 0.075, 0.1, 0.15, 0.2, \ldots, 0.5 \text{ step } 0.05, 0.6, 0.7, 0.8, 0.9\}$; $m \in \{2, 4, 6\}$; $\varepsilon \in \{0.1, 0.01, 0.001\}$; $M = 2^{20}$; $R = 16$. `gauss`: 50 seeds; `population`: exact coefficients, 5 belief seeds; `families`: $k \in \{2, 4, 8\}$, 20 seeds; `lkj`: $k \in \{2, 4, 8\}$, $\eta \in \{1, 4\}$, 200 matrices; `convergence`: $k \in \{2, 8, 32\}$, 5 seeds; `network`: as in Section 4. The budget rule for removing seeds was not triggered.

C Full threshold tables

D Hypothesis details

Decision rules are quoted in substance from Section 6 of the pre-registration. H1a, H2a, H2b (population): deterministic inequality per cell. H1b, H2b ($n = 10^7$), H5a–d: aggregation rule of Section 4. H1c: power law per (type, basis), ≥ 90 % of points within a factor 2 for *supported*, < 50 % for *not supported*. H2c: max / min over m at most 1.5 at ≥ 80 % of points for *supported*, ≤ 50 % for *not supported*. H3a: *supported* if every CI is inside $[-0.05, 0.05]$, *not supported* if any is entirely outside. H3b: same with the ratio in $[0.5, 2]$. H3c: ≥ 80 % for *supported*, < 50 % for

type	basis	k	$d_{\text{denom}} \geq 1\,\%$	$d_{\text{denom}} \geq 5\,\%$	onset, $n = 10^7$	onset, pop.	undefined from
SUM	pair	2	0.192 (0.192–0.193)	0.359 (0.359–0.360)	0.435 (0.435–0.436)	0.437	0.450
SUM	pair	4	0.102 (0.101–0.102)	—	0.201 (0.201–0.201)	0.204	0.200
SUM	pair	8	0.067 (0.067–0.067)	—	0.123 (0.123–0.123)	0.129	0.150
SUM	pair	16	0.043 (0.043–0.043)	—	0.059 (0.059–0.060)	0.074	0.075
SUM	pair	32	—	—	0.007 (0.007–0.008)	0.039	0.010
SUM	triple	2	0.085 (0.085–0.086)	—	0.435 (0.435–0.436)	0.437	0.450
SUM	triple	4	0.020 (0.020–0.020)	—	0.085 (0.083–0.087)	0.094	0.100
SUM	triple	8	0.009 (0.009–0.009)	—	0.016 (0.015–0.017)	0.045	0.020
SUM	triple	16	—	—	0.000 (0.000–0.000)	0.023	0.000
SUM	triple	32	—	—	0.000 (0.000–0.000)	0.011	0.000
CONTRAST	pair	2	0.192 (0.192–0.192)	0.388 (0.387–0.388)	0.435 (0.435–0.435)	0.437	0.450
CONTRAST	pair	4	0.122 (0.121–0.122)	—	0.201 (0.201–0.201)	0.204	0.200
CONTRAST	pair	8	0.084 (0.084–0.084)	—	0.123 (0.123–0.123)	0.129	0.150
CONTRAST	pair	16	—	—	0.059 (0.059–0.060)	0.074	0.075
CONTRAST	pair	32	—	—	0.007 (0.007–0.008)	0.039	0.010
CONTRAST	triple	2	0.065 (0.065–0.065)	0.235 (0.235–0.235)	0.435 (0.435–0.435)	0.437	0.450
CONTRAST	triple	4	0.054 (0.054–0.054)	—	0.084 (0.082–0.086)	0.094	0.100
CONTRAST	triple	8	—	—	0.015 (0.013–0.018)	0.045	0.020
CONTRAST	triple	16	—	—	0.000 (0.000–0.000)	0.023	0.000
CONTRAST	triple	32	—	—	0.000 (0.000–0.000)	0.011	0.000

Table 6: All thresholds for S2, $m = 4$, with 95 % bootstrap CIs. "—" as in Table 2.

not supported. H4a: $\geq 90\,\%$ for *supported*, $< 50\,\%$ for *not supported*. H4b: every violation is listed in `results2/summary/convergence_violations.csv`.

E Numerics

$\min D$ uses 2,000 starts (corners, edge midpoints, the 256 lowest-D Monte Carlo samples of S2 and S4, and uniform random points), projected Adam for 400 steps, cosine learning rate from 0.005 to 0.0001, projected onto the support for Beta beliefs, and takes the minimum with the Monte Carlo minimum. These settings were chosen on $d = 3$ coefficients against brute-force grids, not on production data. Random numbers come from PCG64 with `SeedSequence(12345,...)`, with common random numbers across conditions. Bootstraps use seed 12345.

F Validation

All code-validation tests (M1–M12, M14) pass on an A100 at a single commit (`d06df20`, job 3497120). M13 is not a code test: it is hypothesis H4b on production data and appears in Section 5 as *not supported*.

For instance, M6 checks Eq. (5) in sample: the maximum residual is 2.22×10^{-16} over 36 defined cases. M9 finds a largest difference of 8.2×10^{-5} between the optimiser and reference minima. M4 reproduces 2088 TV values of the $d = 3$ study, with worst difference over tolerance 0.33.

Note on M7. The Student-t part of M7 was rewritten after production. The earlier check compared $1 - F$ with SciPy [55] and could not resolve tails near $F = 1$; it failed on CPU and GPU alike with SciPy 1.13.1. The reference is now 50-digit mpmath [54]: relative error 6.2×10^{-15} in the lower tail, absolute error 2.8×10^{-16} on the upper side, and relative error 1.6×10^{-14} for the quantile's tail probability. No production code changed; the test changed.

G Optimiser settings

Adam: learning rate 0.003, cosine decay to 0, at most 2000 full-batch steps, validation every 50 steps, patience 10. L-BFGS: strong Wolfe line search, learning rate 1, 20 iterations per step,

type	basis	k	$\lvert\text{relvar}\rvert \geq 5\,\%$	$\lvert\text{relvar}\rvert \geq 20\,\%$	$\min D$ at $\rho_S = 0$
SUM	pair	2	0.283 (0.282–0.283)	—	0.992
SUM	pair	4	0.147 (0.146–0.147)	—	0.965
SUM	pair	8	0.095 (0.095–0.095)	—	0.901
SUM	pair	16	—	—	0.708
SUM	pair	32	—	—	0.254
SUM	triple	2	0.246 (0.245–0.246)	—	0.992
SUM	triple	4	0.048 (0.048–0.048)	—	0.866
SUM	triple	8	—	—	0.266
SUM	triple	16	—	—	-1.948
SUM	triple	32	—	—	-10.066
CONTRAST	pair	2	0.299 (0.299–0.299)	—	0.992
CONTRAST	pair	4	—	—	0.966
CONTRAST	pair	8	—	—	0.895
CONTRAST	pair	16	—	—	0.711
CONTRAST	pair	32	—	—	0.253
CONTRAST	triple	2	0.139 (0.139–0.139)	—	0.992
CONTRAST	triple	4	—	—	0.863
CONTRAST	triple	8	—	—	0.250
CONTRAST	triple	16	—	—	-2.068
CONTRAST	triple	32	—	—	-10.202

Table 7: Thresholds of the relative variance error (S2, $m = 4$) and $\min D$ at $\rho_S = 0$. "—" as in Table 2.

type	k	S2	S3a	S3b	S3c	S3d	S4
SUM	2	0.192 / 0.45	— / 0.45	— / 0.45	0.217 / 0.45	— / 0.60	— / 0.45
SUM	4	0.102 / 0.20	— / 0.25	0.039 / 0.25	0.125 / 0.20	— / 0.35	— / 0.20
SUM	8	0.067 / 0.15	— / 0.15	0.014 / 0.20	— / 0.15	— / 0.35	— / 0.15
SUM	16	0.043 / 0.075	— / 0.15	0.009 / 0.90	— / 0.075	— / —	— / 0.075
SUM	32	— / 0.010	0.054 / —	0.027 / —	— / 0.050	— / —	— / 0.010
CONTRAST	2	0.192 / 0.45	— / 0.45	— / 0.45	0.102 / 0.45	— / 0.60	— / 0.45
CONTRAST	4	0.122 / 0.20	— / 0.25	— / 0.25	0.177 / 0.25	— / 0.35	— / 0.20
CONTRAST	8	0.084 / 0.15	— / 0.15	— / 0.20	— / 0.15	— / 0.35	— / 0.15
CONTRAST	16	— / 0.075	— / 0.15	— / 0.90	— / 0.075	— / —	— / 0.075
CONTRAST	32	— / 0.010	— / —	— / —	— / 0.050	— / —	— / 0.010

Table 8: 1 % threshold of d_{denom} and undefined onset (shown as threshold / onset) for all beliefs, pair basis, $m = 4$. "—" in the onset means p_{EU} stayed defined up to $\rho_S = 0.9$; only then does "—" in the threshold mean that the level was not reached in the range.

history 20, at most 50 steps, validation every step, patience 10. Best validation state kept. $\varepsilon = 0.01$ in the likelihood loss. The value loss uses rows $i \in \{0, 1\}$; an all-rows variant is identical because its extra gradients vanish exactly (test M14). The e2e loss uses 2^{16} samples per belief per split. The e2e run for depth L trains the first L neurons and evaluates them followed by the untrained remainder.

H Full energy tables

I The earlier three-input study

This separate study is not pre-registered and motivated the present design; it uses $d = 3$ variables, that is $k = 2$, with $m = 4$. With a flat belief at $\rho_S = 0.2$, d_{denom} was 2.0 % (SUM) and 4.1 % (CONTRAST), and for a narrow belief at $\rho_S = 0.3$ it was 0.0005 and 0.0039. Over 145 rows where p_{EU} was undefined, p_{CD} was closer to p_{G} than p_{EC} in 32 %. A Taylor-corrected denominator with $K = 0, 1, 2$ terms (SUM, S2) gave TV to p_{EU} of 0.0116, 0.0022, 0.0009 at $\rho_S = 0.1$, and 0.0353, 0.0412, 0.0758 at $\rho_S = 0.4$: it helps at small ρ_S and hurts once the series condition fails. These

basis	k	$\lvert B_D\rvert$	σ_c	$1 - \min D$	ratio	χ_D	$\sqrt{\lvert B_D\rvert/n}$
pair	2	24	0.008	0.008	1.0	1.52×10^{-3}	1.55×10^{-3}
pair	4	112	0.019	0.035	1.8	3.32×10^{-3}	3.35×10^{-3}
pair	8	480	0.040	0.099	2.5	6.87×10^{-3}	6.93×10^{-3}
pair	16	1,984	0.083	0.292	3.5	1.41×10^{-2}	1.41×10^{-2}
pair	32	8,064	0.169	0.746	4.4	2.84×10^{-2}	2.84×10^{-2}
triple	4	368	0.077	0.134	1.8	5.99×10^{-3}	6.07×10^{-3}
triple	8	4,064	0.281	0.734	2.6	2.01×10^{-2}	2.02×10^{-2}
triple	16	37,824	0.884	2.948	3.3	6.15×10^{-2}	6.15×10^{-2}
triple	32	325,504	2.624	11.066	4.2	1.80×10^{-1}	1.80×10^{-1}

Table 9: Noise scale at $\rho_S = 0$, $m = 4$, $n = 10^7$, type SUM, S2 (post hoc). σ_c from Eq. (10); "ratio" is $(1 - \min D)/\sigma_c$. The triple basis at $k = 2$ is omitted because it equals the pair basis for D.

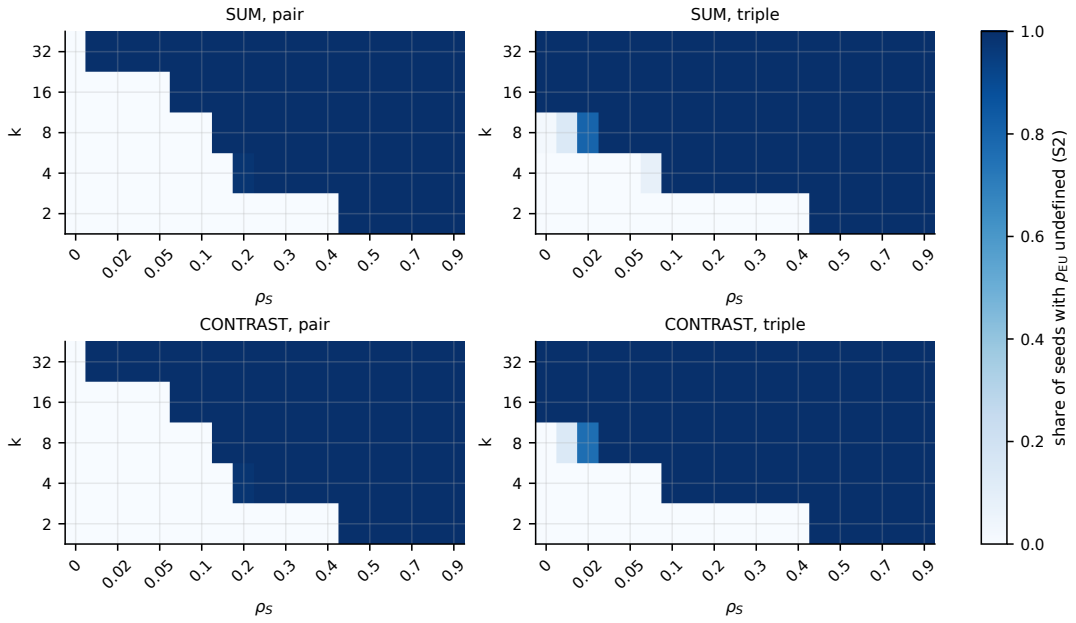


Figure 8: Share of seeds with p_{EU} undefined under S2, $m = 4$, $n = 10^7$. The ρ_S axis is categorical (grid values, unevenly spaced). Dark regions are where d_{denom} cannot be measured.

values come from a different setup and should not be compared directly with the main results.

J Reproducibility

Commits. Pre-registration `6a98646`. Production ran across 5 commits (Table 19): between `6a98646` and `3dd5707`, the only change in the `gpu/` and `network/` code is in `gpu/copulas.py` (Student-t evaluated on the device); the other changes are in scheduling and cost files. All t jobs ran after that change. Validation used `d06df20`, and the energy benchmark `2d26201`.

Seeds. Listed in Section 4 and Appendix B; the pilot seed index 1000 is excluded. **Hardware.** NVIDIA A100-SXM4-40GB, driver 595.91.07, float64, TF32 disabled, deterministic algorithms. **Software.** PyTorch [46] 2.4.0+cu121, NumPy 1.26.4, SciPy 1.13.1.

Regeneration. The repository README gives the commands for the study. For this paper, four scripts in `paper/` do the rest: `posthoc.py` computes post hoc statistics from the merged results, `make_numbers.py` generates every number and data table, `check_numbers.py` re-derives them from source, and `check_identity.py` verifies Section 3 numerically. Every result in the text comes from these generated macros and tables.

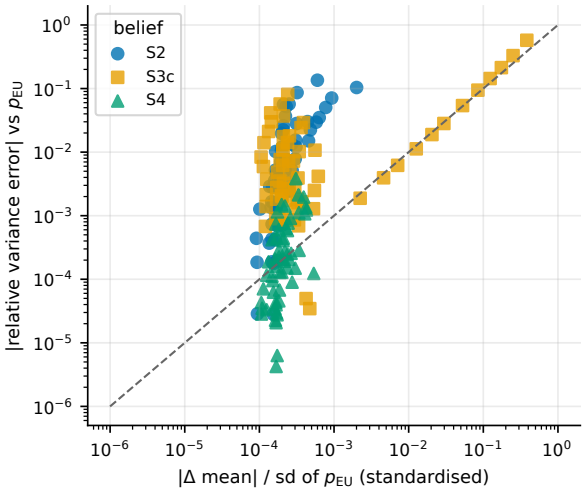


Figure 9: Absolute relative variance error against standardised mean error, both relative to p_{EU}, pair basis, $m = 4$, $\varepsilon = 0.01$, all k, types and $\rho_S > 0$ where defined; seed medians. Points above the dashed line have larger variance error than mean error.

belief	type	ρ_S	d_{denom}	d_{neg}	$d_{\mathrm{model,uncal}}$	$d_{\mathrm{model,cal}}$	d_{total}
S2	SUM	0.1	0.0025	0.0111	0.0131	0.0241	0.0106
S2	SUM	0.2	0.0108	0.0076	0.0287	0.0358	0.0180
S2	SUM	0.4	0.0736	0.0376	0.1002	0.0627	0.0265
S2	CONTRAST	0.1	0.0025	0.0123	0.0129	0.0026	0.0154
S2	CONTRAST	0.2	0.0108	0.0095	0.0267	0.0195	0.0374
S2	CONTRAST	0.4	0.0547	0.0238	0.0397	0.0617	0.0906
S3c	SUM	0.1	0.0034	0.0064	0.0399	0.0445	0.0368
S3c	SUM	0.2	0.0087	0.0064	0.0372	0.0415	0.0294
S3c	SUM	0.4	0.0355	0.0061	0.0485	0.0438	0.0171
S3c	CONTRAST	0.1	0.0097	0.0391	0.0650	0.0673	0.0710
S3c	CONTRAST	0.2	0.0262	0.0551	0.0854	0.0971	0.1022
S3c	CONTRAST	0.4	0.1030	0.1116	0.0985	0.1371	0.1526
S4	SUM	0.1	0.0001	0.0120	0.0003	0.0119	0.0003
S4	SUM	0.2	0.0004	0.0103	0.0004	0.0100	0.0003
S4	SUM	0.4	0.0003	0.0037	0.0004	0.0037	0.0002
S4	CONTRAST	0.1	0.0001	0.0118	0.0003	0.0118	0.0003
S4	CONTRAST	0.2	0.0004	0.0104	0.0004	0.0101	0.0003
S4	CONTRAST	0.4	0.0003	0.0044	0.0004	0.0045	0.0004

Table 10: Error components at $k = 2$, pair basis, $m = 4$, $n = 10^7$, $\varepsilon = 0.01$, seed medians.

type	basis	points	β	C	within 2×	cell verdict
CONTRAST	pair	33	1.59	0.040	0.52	inconclusive
CONTRAST	triple	23	1.00	0.083	0.48	not supported
SUM	pair	33	1.76	0.070	0.55	inconclusive
SUM	triple	23	0.88	0.107	1.00	supported

Table 11: H1c fits $d_{\mathrm{denom}} = C\chi_D^{\beta}$ on population coefficients, S2, $m = 4$. Cell verdicts follow the H1c rule.

basis	k	$m = 2$	$m = 4$	$m = 6$
pair	2	0.545	0.437	0.398
triple	2	0.545	0.437	0.398
pair	4	0.261	0.204	0.168
triple	4	0.199	0.094	0.070
pair	8	0.158	0.129	0.116
triple	8	0.087	0.045	0.035
pair	16	0.090	0.074	0.068
triple	16	0.041	0.023	0.018
pair	32	0.049	0.039	0.038
triple	32	0.021	0.011	0.009

Table 12: Population negativity onset by degree (H2a), type SUM.

k	η	defined	drawn	within 2×, pair	within 2×, triple
2	1	108	200	38 %	49 %
2	4	165	200	39 %	24 %

Table 13: H3c by k and η. "defined" counts matrices with p_{EU} defined (pair basis, S2, $m = 4$). Shares are among defined matrices.

variant	paired values	violations	median change	max change	worst ratio to tol.
$M \times 2$	291,249	0	1.9×10^{-5}	4.6×10^{-1}	0.86
$n \times 2$	291,258	5,460	4.6×10^{-5}	8.3×10^{0}	36.21
s-nodes ×2	291,276	0	2.7×10^{-9}	3.7×10^{-5}	0.00
x-nodes ×2	291,276	0	4.0×10^{-10}	6.4×10^{-6}	0.00

Table 14: Convergence test H4b on the `convergence` subset ($k \in \{2, 8, 32\}$, 5 seeds). Paired values are reported TVs before and after doubling one setting; a violation is a change of at least $\max(3\,\mathrm{SE}, 10^{-3})$. The final column is the largest ratio of change to tolerance.

loss	optimiser	kept initial	coef. dist.	test loss	test loss down	$\Delta P(D < 0)$	$\Delta \min D$
value	Adam	23 %	0.132	−0.3 %	52 %	0.005	−0.47
value	L-BFGS	4 %	0.131	−1.1 %	82 %	0.000	−0.34
NLL	Adam	0 %	0.649	−10.9 %	100 %	−0.055	−7.35
NLL	L-BFGS	0 %	0.660	−11.1 %	100 %	−0.043	−8.06
e2e, $L = 1$	Adam	3 %	0.432	−97.0 %	96 %	0.109	−13.67
e2e, $L = 1$	L-BFGS	5 %	0.216	−97.7 %	94 %	0.060	−1.21
e2e, $L = 2$	Adam	10 %	0.167	−91.7 %	88 %	0.035	−2.12
e2e, $L = 2$	L-BFGS	9 %	0.059	−91.8 %	88 %	0.001	−0.04
e2e, $L = 3$	Adam	19 %	0.154	−52.2 %	71 %	0.033	−1.34
e2e, $L = 3$	L-BFGS	18 %	0.045	−50.2 %	71 %	0.001	−0.02
e2e, $L = 4$	Adam	54 %	0.000	0.0 %	31 %	0.003	−0.00
e2e, $L = 4$	L-BFGS	50 %	0.000	0.0 %	33 %	0.000	0.00

Table 15: Training runs from direct estimates, medians over runs (post hoc). "kept initial": share of runs whose best validation state is the initial one. "coef. dist.": coefficient distance moved. "test loss": relative change of the training loss on the test split, and the share of runs with a decrease. $\Delta P(D < 0)$ is taken where $P(D < 0)$ was positive. $\Delta \min D$ is the change in $\min D$.

test	what it checks	result
M1	orthonormality of the basis, explicit $f_1, \ldots, f_4$	pass
M2	index families, coefficient counts, block vs dense estimator	pass
M3	estimator vs population moments, SE scaling, Gauss–Hermite convergence	pass
M4	reproduces the $d = 3$ study (TV values, flags)	pass
M5	flat-D control: $p_{\mathrm{CD}} = p_{\mathrm{EU}}$	pass
M6	covariance identity in sample	pass
M7	integrals of all densities; copula samplers and conditionals; Student-t CDF and quantile against 50-digit mpmath	pass
M8	TV in s vs x; grid refinement with spiky p_{G}	pass
M9	min-D optimiser vs brute-force grids and closed forms	pass
M10	positive definiteness of all correlation matrices; LKJ moments	pass
M11	bitwise reproducibility; kill and resume of a job	pass
M12	seed variance matches estimated SE; common random numbers	pass
M13	convergence under doubling = hypothesis H4b, on production data	**fail**
M14	chain: $L = 1$ vs engine, closed form vs MC, autograd vs finite differences, teacher Markov property	pass

Table 16: Validation tests. M13 "fail" is the H4b outcome.

layer	k	B	params	ms	J	J net	W	MHz	counter
HCR-CAL	2	256	3,904	4.52	1.24	0.983	275	1410	0.997
HCR-CD	2	256	3,904	0.147	0.00891	0.00043	61	1095	1.017
KAN	2	256	4,158	1.06	0.0655	0.00319	62	1095	0.995
MLP	2	256	3,883	0.0657	0.00399	0.000234	62	1095	0.997
HCR-CAL	2	16,384	3,904	285	82.5	66.1	290	1410	1.003
HCR-CD	2	16,384	3,904	0.178	0.0457	0.0354	256	1410	0.980
KAN	2	16,384	4,158	1.07	0.0946	0.0338	92	1260	0.992
MLP	2	16,384	3,883	0.145	0.0165	0.00814	114	1410	0.993
HCR-CAL	8	256	39,232	4.87	1.37	1.09	283	1410	1.009
HCR-CD	8	256	39,232	0.154	0.0113	0.00231	74	1185	0.993
KAN	8	256	39,528	1.08	0.0688	0.00538	64	1095	1.017
MLP	8	256	39,265	0.0685	0.00498	0.000946	73	1170	0.998
HCR-CAL	8	16,384	39,232	308	88.7	71	289	1410	0.992
HCR-CD	8	16,384	39,232	0.484	0.137	0.109	283	1410	1.011
KAN	8	16,384	39,528	1.91	0.482	0.372	253	1410	1.002
MLP	8	16,384	39,265	0.289	0.0682	0.0515	236	1410	1.003
HCR-CAL	32	256	549,184	10.4	3.13	2.53	300	1410	1.003
HCR-CD	32	256	549,184	0.152	0.0342	0.0254	226	1410	0.994
KAN	32	256	549,504	1.06	0.115	0.0528	108	1410	0.988
MLP	32	256	549,181	0.0713	0.0127	0.00859	179	1410	1.004
HCR-CAL	32	16,384	549,184	663	202	164	305	1410	1.000
HCR-CD	32	16,384	549,184	5.93	1.89	1.55	318	1410	1.004
KAN	32	16,384	549,504	16.9	4.77	3.79	282	1410	0.996
MLP	32	16,384	549,181	1.93	0.682	0.571	353	1410	1.009

Table 17: Float32 forward pass: median over 90 repeats (3 runs) of time, energy, energy net of idle, power, SM clock, and the ratio of the energy counter to sampled energy.

layer	k	B	params	ms	J	J net	W	MHz	counter
HCR-CAL	2	256	3,904	8.76	2.49	1.97	284	1410	1.000
HCR-CD	2	256	3,904	0.143	0.00908	0.000623	63	1095	0.980
KAN	2	256	4,158	1.03	0.0628	0.00257	61	1095	0.989
MLP	2	256	3,883	0.0693	0.00432	0.000278	63	1095	0.989
HCR-CAL	2	16,384	3,904	557	159	126	285	1410	0.989
HCR-CD	2	16,384	3,904	0.328	0.083	0.0638	253	1410	0.983
KAN	2	16,384	4,158	1.04	0.13	0.0697	126	1410	0.984
MLP	2	16,384	3,883	0.0667	0.0124	0.00851	186	1410	1.006
HCR-CAL	8	256	39,232	9.11	2.56	2.03	281	1410	0.983
HCR-CD	8	256	39,232	0.147	0.0111	0.00251	76	1260	1.001
KAN	8	256	39,528	1.01	0.0653	0.00515	63	1095	1.003
MLP	8	256	39,265	0.0691	0.0049	0.000862	71	1095	0.990
HCR-CAL	8	16,384	39,232	580	168	134	289	1410	0.989
HCR-CD	8	16,384	39,232	0.666	0.203	0.164	304	1410	1.003
KAN	8	16,384	39,528	3.32	0.829	0.634	250	1410	1.001
MLP	8	16,384	39,265	0.297	0.105	0.0879	355	1410	1.005
HCR-CAL	32	256	549,184	15	4.69	3.81	313	1410	1.009
HCR-CD	32	256	549,184	0.241	0.0527	0.0386	219	1410	0.996
KAN	32	256	549,504	1.03	0.146	0.0872	145	1410	0.997
MLP	32	256	549,181	0.0837	0.0185	0.0136	223	1410	1.006
HCR-CAL	32	16,384	549,184	955	300	243	313	1410	0.993
HCR-CD	32	16,384	549,184	7.41	2.15	1.71	289	1410	1.000
KAN	32	16,384	549,504	31.1	8.18	6.35	263	1410	0.999
MLP	32	16,384	549,181	2.8	1.11	0.948	398	1388	1.002

Table 18: Float64, as Table 17.

experiment	jobs	GPU-h	`6a98646`	`8addd92`	`9bac5d5`	`b8e5341`	`3dd5707`
`gauss`	1,000	66.0	6	137	857	—	—
`population`	100	11.6	—	1	99	—	—
`convergence`	240	23.2	—	—	240	—	—
`families`	1,080	42.5	—	—	1	42	1,037
`lkj`	240	2.8	—	2	24	1	213
`network`	240	30.6	—	—	—	—	240
total	2,900	176.7	6	140	1,221	43	1,490

Table 19: Jobs and GPU-hours by experiment, and jobs by commit.